Prepared for referee review

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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This response addresses the 11 October 2026 advisory report on R67. Revision R70 retains the original Neural Bellman Operators paper and integrates the separately frozen R69 execution, new deployment and cost-inference results, independent reconstruction and a point-by-point response.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R70*

We thank the referee for identifying the remaining connections between the constructive operator, the trained selector, verified work and the economic comparison. The controlling advisory report, dated 11 October 2026, is commit 4b27b56f4514f6c1d77b69fe707f7202f8b06b19; it reviews R67 manuscript commit eacd3e217ae1e2e9ea6e154b48da7d103159d051. This response concerns that report, not an earlier R54 review or an editorial decision by the Econometric Society.

We have revised the original **Neural Bellman Operators** paper. Its title, own-future continuation, constructive neural realization, feasible witness mechanism, original Bellman optimum, investment primitives and broader historical applications are retained. Rather than replacing the topic with a generic verifier paper, the revision completes the connections among those objects. Every inherited active mathematical statement is retained verbatim. Proofs are retained verbatim in the article or supplement. The full preceding empirical article, supplement and complete development remain unchanged companions, with a source and label map.

The revision integrates the separately frozen R69 execution at commit 2c07e464e81930f14b3122054704731ec5a139ff. Its 188-process catalogue was specified before execution. R70 does not relabel those processes as new experiments, add samples after inspecting their signs, or treat duplicate timing streams as independent information. The incomplete R68 upload remains on its own branch and is not scientific evidence.

Three additions are central. First, the fixed-feature certificate-aware readout is now executed alongside equal-information and equal-size Bernstein and label-loss controls. Second, a new deployment theorem extends a uniform independent-validation work account to predictable readout selection and dependent contexts under an explicit drift premise and complete descendant costs. Third, a Bellman-centered estimator identifies the original expected policy-cost difference at a regret-scale range, with its optimal-value integration explicitly enclosed and charged. All twelve new scalar contrasts meet the prospectively declared equivalence margin, although none identifies strict superiority.

The active reading set is the article, technical supplement and this response. The new sections are **Policy-cost inference at the scale of verified regret**, **From validated readouts to complete deployment**, and **Training, reuse, and policy-cost precision**. The release audit binds ordinary sources, full numerical records, tests, independent path reconstruction, compiled documents and a clean-archive rebuild. A successful numerical build is not represented as external mathematical acceptance.

1.2. *B1 and M1: The contribution of prediction within the original NBO method*

The referee correctly separates the verifier's correctness from the value of a particular predictor. The revised introduction presents the two constructive implementations of the original NBO program: the witness-preserving own-future continuation and recursive original-Bellman queries. The learned readout remains an untrusted proposal to the latter. It cannot change the lower certificate, the feasibility contract or the original-optimum guarantee.

The new comparison purchases identical training states, action labels and boundary endpoints for the learned scalar arms. Certificate-aware hat–ReLU and degree-four Bernstein readouts each have exactly $5d$ box-constrained coefficients, the same certificate objective and penalty, and the same optimization budget. A label-loss readout uses the same hat features with a different objective. The quadratic arm shares information but retains its different parameter count. Adaptive search and an

upper-chord minimizer remain explicit prediction-free controls and are not charged unused training.

This identifies the representation, objective and verifier separately. A hat function has both a conventional piecewise-linear interpretation and an exact ReLU realization; those encodings are not counted as two independent methods. The positive contribution is an executed constructive training–certificate–policy chain, not a presumption that a ReLU label establishes a computational advantage. Cold and reuse results remain unfavorable to universal neural dominance and are reported as such.

1.3. *B5 and M2: Execute the certificate-aware optimization result*

R67 proved the readout proposition but its main fitted comparison used action-label training. The new frozen catalogue explicitly optimizes the stated certificate-aware squared-surplus objective. Each returned iterate is quantized and then evaluated with exact rational contexts, features and coefficients. The publication replay recomputes the achieved objective, box dual gap and associated training-work bound. There are twelve distinct rational readout certificates, represented in 108 separately charged service/date records.

The supplementary optimization table reports those achieved quantities. An optimizer's success message is not substituted for the dual gap. This is the box-constrained fixed-feature problem actually covered by the proposition, not unrestricted hidden-layer training. The original action-label experiment remains unchanged and is not retrospectively described as certificate-aware fitting.

1.4. *B5 and M3: Independent validation and dependent deployment*

The new study includes sixteen independent-validation services, producing twenty-four task–date–method–seed records of 256 contexts each. The contexts are predictor-independent and disjoint from fitting. Each coordinate is drawn uniformly as U/2^16, with integer U from zero to 2^16-1; this finite law differs from both the full twenty-four-bit acquisition grid and the continuous law used for policy-cost inference. Shared contexts across methods are retained as shared

data, not multiplied into new independent observations. The recorded recovery is a complete-action mesh with a declared root-query cap. Every failed screen executes that recovery and its descendants. The replay verifies the saved action's exact strong-convexity chord, the actual recovery count, the pointwise work-loss domination and the returned gap.

The validation upper bound is recomputed from an exact sample sum and an outward concentration radius. The main article and supplement distinguish root work from descendant work and context acquisition. They also state that this validation law and mesh fallback are not automatically the law and fallback of the adaptive deployment service.

The new **Validated work under predictable deployment** theorem makes the missing bridge explicit. A readout may be selected using validation and the prior deployment history. A total-variation allowance bounds the conditional context-law change; a conditional bounded-difference argument controls realized work; and deterministic endpoint-work caps charge recursive descendants and numerical recovery. The proof requires neither independent deployment contexts nor a hidden minimizing action. A model hash is not treated as a context-law certificate. The R69 deployment records do not purport to estimate an unverified drift allowance; they supply the separate measured-work comparison.

1.5. *B2 and M8: Complete costs and timing resolution*

The cold-start table includes all eighty complete processes and all prescribed repetitions. The parent clock starts before launch and ends after process exit and flushed output. It includes imports, purchased training information, fitting, prediction, all Bellman queries, interval path evaluation, serialization and shutdown. The group record separately includes parent receipt bookkeeping. Query counts do not silently exclude descendants, and training counts remain separate from the policy workload.

The hat–ReLU median is approximately 0.465 seconds against 0.365 for adaptive search in the small task, and 1.618 against 0.666 in the larger task. The latter workload has 2,594 policy-query evaluations for both rules, before separately charged

training. These observations do not support a cold-start neural work advantage. They instead locate the overhead that the trained readout must repay.

Four repetitions, complete ranges, operation counts, resident memory, permitted CPU and available frequency/governor metadata are retained. Frequency was not locked by the experiment. Closely spaced clock values are descriptive and are not promoted to statistically identified runtime rankings. Failed or warned fits would remain in the catalogue; the earlier forty-eight neural-fitting warnings remain in the preceding study.

1.6. *B3 and M4: A longer, prospective reuse window*

The reuse catalogue expands the observation window to sixty-four predetermined blocks and uses four process repetitions per seed and task. Each process trains and serializes once, loads and validates its saved model at each block, and rejects a deliberately stale acquisition identifier at block index 16. The same adaptive verifier handles that block. All related costs are paid.

The prospective stable-crossing criterion requires a cumulative crossing by index 48 that persists through index 63. Three of the eight small-task hat–ReLU processes meet that criterion; four have lower final complete times than the matched adaptive process. Bernstein and quadratic rules also have small-task crossings. None of the learned arms wins the larger-task complete reuse comparison.

Every matched repetition and crossing is retained, including a slow adaptive repetition that affects the small-task comparisons. The whole observed time range is reported. A favorable pairing is not treated as a population amortization theorem, and the sixty-four-block result is not extrapolated to later blocks. The two scalar task families remain the declared tasks; the revision does not claim a broad multi-economy reuse frontier that has not been executed.

1.7. *B4 and M5: Direct cost inference at a meaningful declared precision*

The new estimator addresses the width problem without increasing sample size after inspecting results. For a frozen feasible policy, the discounted sum of optimal Bellman advantages along its actual trajectory has expectation equal to its policy

loss. Subtracting two such sums cancels the expected initial optimum under the common initial law. Their expected difference is therefore the original expected policy-cost difference. Local regret bounds control the observation range.

The **Bellman-centered cost comparison** theorem proves this identity, its range and a simultaneous endpoint confidence interval. Its proof does not assert a pathwise cash-flow identity, rely on an assumed-correct fitted critic, or compare two loss upper bounds as though they were policy costs. The actual returned action is fixed before a separately paid tighter optimal-value evaluation. Continuous-bin uncertainty, all compatible acquired actions, quadrature, action coverage and outward arithmetic remain in that calculation.

The frozen design uses 1,024 new common-path rows for each scalar task, evaluation tolerance equal to the policy target divided by 32, and a loss margin of $1/256$. Numerical and sampling half-width allocations are each $1/512$. All twelve paired contrasts and both allocations are reported. The four methods include the unchanged original R67 ReLU controller loaded from its pinned model, not a retrained replacement.

All twelve intervals lie within the declared equivalence margin, while all still contain zero. This establishes two-sided equivalence at that normalized-loss margin, not equality or strict superiority. For adaptive-minus-hat–ReLU, the centered intervals are approximately $[-0.00068263, 0.00070764]$ and $[-0.00160291, 0.00160253]$, rounded outward. Their widths are about 0.00139 and 0.00321, compared with about 0.423 and 0.529 for ordinary cash-cost intervals on the same new rows. The target has not changed.

The inference service is not free: the complete shared processes take approximately 9.03 and 408.63 seconds. Conditional Bellman integration dominates the numerical width; bin propagation is much smaller. Raw pre-intersection components are displayed as diagnostics and not misrepresented as an additive decomposition of the final intersected width. The two new tasks contain 2,048 common-path rows, not that number multiplied by methods, pairs or publication replays. Combining centered and ordinary cash-cost statistical families requires the corresponding union bound.

1.8. *B6 and M6: Trained multivariate-control comparisons*

The new vector catalogue contains simplicial, largest-edge uniform, two-output hat and two-output quadratic rules under the same triangular certificate. The learned rules purchase the same twelve teacher states per nonterminal date and use the same feasible rounding contract. Both original continuous innovations, their coupling to the states and the nonseparable action cost are retained. Ordinary vector services cover dimensions two and eight and horizons two and three, with two process repetitions.

The two additional horizon-four services compare simplicial search with transfer of a three-date hat readout. This is explicitly a transfer comparison, not four-date supervised training. Its one path is a workload diagnostic, not a population cost estimate. The vector training objective is action-label loss; it is not relabeled as the scalar convex certificate-aware objective.

The vector table reports all complete clocks, descendant queries and peak memory, and the original severe horizon-four growth remains in the unchanged R67 records. Derivative branch-and-bound is not an executed arm of the new catalogue. The added result is a real trained vector comparison within the original economy, not a claim that the long-horizon tree cost has disappeared or that a vector expected-cost ranking has been identified.

1.9. *B7 and M7: Independent numerical implementation and retained disagreements*

The new independent path implementation imports no NBO numerical kernel. It uses exact rational intervals for the full continuous initial and shock bins, acquisition comparisons, feasible actions, stage and terminal costs, and transitions. Coordinate monotonicity bounds the bilinear transition without invoking the production interval library. Its complete-bin cost enclosure is checked against the archived enclosure. A separate rational midpoint implementation supplies an additional point check, not a replacement for the full-bin calculation.

The replay checks 3,706 physical path archives and their 38,722 recorded rows. Their 736 unique payloads contain 14,242 rows; all receive independent exact full-bin reconstruction, with no unresolved branch in this executed set. Duplicate

timing copies are hash-checked but do not multiply independent arithmetic work or statistical observations. Every stored source and output identity is retained. A separate exact-sum and high-precision calculation checks all twelve final centered intervals from their saved advantage endpoints.

An initial strict validation-predictor replay failed a bitwise equality check for Bernstein. That attempt, code and traceback are preserved. The failure alone did not establish an invalid Bellman enclosure, and a successful run on another processor is not evidence that the initial bitwise discrepancy never occurred. The revised audit checks the saved actual action, its lattice and its exact chord certificate. Any recomputed readout discrepancy must also satisfy the original prediction and rounding allowances. The scientific records are not changed.

The independent scope is explicit. Full-bin sampled cash-path reconstruction and final confidence-interval arithmetic are now independently implemented. Small value and terminal-moment regressions use separate rational formulas. The largest vector optimal-value tree and the complete generation of all optimal-advantage endpoints have not been independently regenerated. No blanket formal-verification claim is made.

1.10. *B8: Return reliability, fitting and numerical identity*

Every one of the 188 new processes returned, but this is not a population training-success probability. The verifier's safety does not depend on a predictor converging. Achieved objective and dual-gap certificates replace reliance on optimization flags for the fixed-feature result. All original seeds, warnings, repetitions and failed historical attempts remain distinguishable.

The publication also distinguishes parameter identity from numerical callable-policy identity. Readout arrays are checked across the relevant inference and service records. That check is necessary, but by itself does not prove all-state equality on different arithmetic runtimes. The purchase proposition requires the same numerical policy or an additional valid implementation-transfer account. A finite successful replay or matching metadata is not silently substituted for that premise.

1.11. *B9 and M9: An operational economic decision*

The new precision result makes a previously unresolved quality margin assessable for actual complete policies under the original law. The deployment proposition then states the full price region for replacing a benchmark controller. For sixty-four blocks of eight trajectories with one predetermined adaptive-fallback block, the expected candidate-minus-benchmark policy cost is 504 times the normal-policy cost difference. The proposition combines that simultaneous interval with complete work, the value of a normalized loss unit and the installation fee.

The supplementary table reports matched observed savings and zero-fee critical price ratios, including all cases with no positive saving. Its interpretation is conditional on numerical policy identity and the declared work boundary. It does not turn a favorable prefix clock into a completed service, or a timing range into a future-latency confidence interval.

The investment primitives and loss margin remain normalized theoretical quantities. No installation fee or monetary welfare magnitude is estimated. This preserves the original economic purchase question while separating a proved eligibility conclusion, observed complete work and the purchaser's price inputs. Equivalence at a margin is a substantive precision result, but it is not strict neural economic superiority.

1.12. *B10 and M10: Scope, style and preservation of the original paper*

The active article follows one theorem–implementation–evidence chain. The original NBO operator and primitive regularity lead to recursive certification, training, centered inference, complete deployment and the executed comparison. Detailed proofs and records are placed in the technical supplement rather than appended indiscriminately to the main text. All relocated proofs are preserved verbatim and all inherited active mathematical statements are checked against their original bytes.

The complete R67 article and supplement are unchanged companion documents, retaining their full empirical sections, conventional winners, unresolved ordinary cost intervals, fitting warnings and interval-accounting counterexample. The broader diffusion, recursive-preference, temporal-self and game results remain

in the complete development under their original assumptions. They are not silently claimed as consequences of the convex expected-cost query theorem.

The paper is therefore revised in place as **Neural Bellman Operators**, not replaced by a different topic or by a lower-ambition generic framework. The new manuscript makes the original constructive program more operational while keeping each empirical and mathematical assertion tied to its actual evidence and hypotheses.

1.13. *Reproduction and renewed review*

The release verifies the original R69 source freeze and all 188 process receipts, checks every recorded-file hash, replays achieved training and validation certificates, independently reconstructs the unique continuous-bin cash paths and final inference arithmetic, runs inherited and new regression tests, and compiles ordinary article, supplement and response sources. A clean-archive rebuild removes generated current PDFs and tables before reproducing them. The final manifest records the exact tested sources, preserved predecessors, outputs and scope of each check.

The submitted positive results are the executed constructive readout comparison, independent validation, a predictable-deployment work theorem with full descendant accounting, a paid centered estimator reaching a declared policy-quality margin, trained vector comparisons and stronger independent full-bin reconstruction. Universal neural complete-work dominance, strict economic superiority, calibrated monetary welfare and independent regeneration of the largest optimal-value trees are not inferred from those results. They remain distinct from the claims submitted for renewed assessment.